

YUXIN YANG

Department of Computer Science, University of Southern California

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EDUCATION

Ph.D. in Computer Science, University of Southern California

Sep 2023 –

- **GPA:** 3.94/4.00
- **Core courses:** Parallel and Distributed Computation (4.0), Advanced Analysis of Algorithms (4.0), Theoretical Machine Learning (4.0), The Science of Large Language Models (4.0)

B.Eng. in Automation, Tsinghua University

Sep 2019 – Jun 2023

- **GPA:** 3.84/4.00

SELECTED AWARDS AND HONORS

- Capital One PhD Fellow 2024 – 2026
- Changtong Scholarship (top 1% in the Department) 2022
- Comprehensive Scholarship of University (top 5% in University) 2020, 2021

PUBLICATIONS AND MANUSCRIPTS

Abdulla Alshabanah, **Yuxin Yang**, Murali Annavaaram. “SAGERec: Sampling and Gating for Enhanced Long-Tail Item Recommendations.” 18th ACM International Conference on Web Search and Data Mining (WSDM), 2026.

Gangda Deng*, **Yuxin Yang***, Ömer Faruk Akgül*, Hanqing Zeng, Yinglong Xia, Rajgopal Kannan, Viktor Prasanna. “Training Diverse Graph Experts for Ensembles: A Systematic Empirical Study.” Preprint, 2025.

Ömer Faruk Akgül*, Feiyu Zhu*, **Yuxin Yang**, Rajgopal Kannan, Viktor Prasanna. “RECIPE-TKG: From Sparse History to Structured Reasoning for LLM-based Temporal Knowledge Graph Completion.” Preprint, 2025.

Yuxin Yang, Hongkuan Zhou, Rajgopal Kannan, Viktor Prasanna. “Towards Ideal Temporal Graph Neural Networks: Evaluations and Conclusions after 10,000 GPU Hours.” 51st International Conference on Very Large Data Bases (VLDB), 2025.

Yuxin Yang, Yitao Liang, and Muhan Zhang. “PA-GNN: Parameter-Adaptive Graph Neural Networks.” Workshop on Dynamic Neural Networks in *the 39th International Conference on Machine Learning (ICML-DyNN-Workshop)*, 2022, oral presentation.

RESEARCH EXPERIENCE

MASC: Multi-Agent Sequential-Parallel Collaboration for RAG

May 2024 – Present

Research Assistant, advisor: Prof. Viktor Prasanna

University of Southern California

In collaboration with Capital One Research

- Designed a unified test-time scaling framework that jointly expands **sequential depth** and **parallel width** for multi-hop Retrieval Augmented Generation (RAG), overcoming the limitations of prior sequential-only or parallel-only methods.
- Built a **multi-agent architecture** with specialized agents and adaptive controllers that dynamically determine reasoning depth and branch width during inference.
- Developed **DPO-based lightweight training** for query rewriting and answer evaluation, improving sub-query diversity, retrieval recall, and reducing premature stopping.
- Conducted a comprehensive scaling study over depth-width configurations; identified practical cost-quality sweet spots under realistic compute budgets.
- Achieved **state-of-the-art multi-hop QA performance** on HotpotQA, 2WikiMultiHopQA, MuSiQue, and Bamboogle using standard RAG setting.

- Designed a unified and fully optimized TGNN comparison framework that isolates and evaluates core modules (neighbor sampling, node memory, aggregation), enabling fair design-space exploration across models.
- Conducted a detailed experimental study to answer three key research questions on efficiency, universality, and module interplay; identified that most-recent sampling, attention aggregation, and node memory yield optimal cost-performance tradeoffs.
- Discovered that dataset repetition patterns fundamentally determine node-memory preference, showing RNN-based memory excels on short-term repetition datasets while static embedding memory excels on long-term repetition datasets.
- Revealed strong diminishing returns in deeper/wider neighbor sampling when node memory is present, leading to practical design recommendations for future TGNN architectures.

- Showed that capturing local patterns of graph datasets improves performance of learning tasks on graphs.
- Designed a trainable node-specific aggregator that learns from node position and features to capture local patterns; Utilized DeepWalk to encode node position.
- Experimentally showed that graph neural networks with an adaptive aggregator can exploit local patterns and improve node classification accuracy by up to 5%.

COMPUTER AND LANGUAGE SKILLS

- **Programming:** Python, C/C++, Javascript
- **Tools:** PyTorch, MATLAB, CMake, LabVIEW, Linux, Git, LaTeX
- **Language:** TOEFL 114/120 (Reading 30, Listening 29, Speaking 25, Writing 30); GRE 327/340+3.5/6.0 (Verbal 157, Quantitative 170, Analytical Writing 3.5)